

A RISK-AWARE AMBIGUITY MANAGER FOR COMPOUND AMBIGUOUS ROBOT COMMANDS

Research Proposal

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Abstract

Natural-language commands are a practical interface for human-robot interaction, but ambiguity still makes reliable execution difficult. In robot command interpretation, ambiguity appears through indirect speech acts, missing references, fuzzy temporal or quantitative language, and safety or capability constraints. Current approaches address these ambiguity types in isolation, but they do not consistently coordinate routing when multiple ambiguity types appear in a single command.

The proposal tests whether a risk-aware ambiguity manager can improve routing decisions for compound ambiguous robot commands. The manager classifies ambiguity type and task risk, then routes commands to one of several strategies: execute, clarify, silently_resolve, face_preserving_rejection, or multi_step handling. Evaluation is conducted on a labelled hybrid benchmark built from AmbiK, the Situational Awareness for Goal Classification (SaGC) dataset, TEACH/TEACH-DA, and IndirectRequests, plus a controlled manual compound-ambiguity extension. Performance is compared against direct Large Language Model (LLM) interpretation and uniform baseline policies (always clarify, always resolve, and degree-based routing), with routing correctness as the primary metric and ambiguity, clarification, intent/slot, and risk-sensitive metrics as supporting measures.

Expected contributions are a routing-focused ambiguity taxonomy, a practical benchmark for compound ambiguity, and evidence on whether explicit type-and-risk-aware coordination improves ambiguity-handling decisions compared with direct LLM parsing and uniform policies in the robot command interpretation layer.

Declaration

I, Mohammed Bangie, hereby declare the contents of this research proposal to be my own work. This proposal is submitted for the degree of Bachelor of Science with Honours in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.

Contents

Preface	
Abstract	i
Declaration	ii
Table of Contents	iii
List of Figures	iv
List of Tables	v
1 Introduction	1
2 Background and Related Work	4
2.1 Natural Language Command Interpretation in Robotics	4
2.2 Large Language Models for Structured Robotic Outputs	4
2.3 Indirect Speech Acts (ISA) in Robot Commands	5
2.4 Clarification and Ambiguity Detection	6
2.5 Fuzzy Temporal and Quantitative Parameters	7
2.6 Risk, Capability, and Transparent Rejection	7
2.7 Synthesis and Research Gap	8
3 Research Methodology	9
3.1 Research Design	9
3.2 Proposed Risk-Aware Ambiguity Manager	9
3.3 Dataset Strategy	11
3.4 Annotation Schema	12
3.5 Baseline Systems	13
3.6 Evaluation Metrics	14
3.7 Experimental Procedure	15
3.8 Failure Analysis	16
3.9 Limitations and Ethical Considerations	16
4 Schedule of Work	17
5 Conclusion	19
References	21

List of Figures

2.1	Interpreting a natural-language imperative requires identifying the speech act (intent), core propositional content (new information), and supplemental propositional content (given knowledge used for reference resolution). Adapted from Sarathy <i>et al.</i> [2025].	5
2.2	The ClarifyVC stage-wise generation flow. The schema-constrained pipeline combines semantic parsing, adversarial construction, and multi-turn evolution to resolve ambiguous commands. Adapted from Zhou <i>et al.</i> [2026].	6
3.1	Proposed risk-aware ambiguity manager. The manager classifies ambiguity type, task risk, and capability status before routing a command to an appropriate handling strategy. Evaluation checks whether the selected route and structured interpretation match the labelled benchmark. . . .	11

List of Tables

2.1	Summary of current capabilities and identified research gaps across the surveyed literature (ISA: Indirect Speech Act).	8
4.1	Semester 2 (August–October) schedule for implementation, evaluation, and write-up.	17

Chapter 1

Introduction

Natural-language instruction is useful for human-robot interaction because non-expert users can use it in everyday terms. The trade-off is ambiguity. A single command can include pragmatic indirection, missing references, vague temporal or quantitative parameters, or implicit safety and feasibility assumptions [Zhang *et al.* 2025; Madureira and Schlangen 2023; Sucker *et al.* 2024; Sucker and Henrich 2025]. In embodied settings, ambiguity is not just a language nuisance: misinterpretation can cause task failure, unsafe decisions, avoidable clarification loops, and reduced user trust [Scheutz *et al.* 2022; Zhang *et al.* 2025]. For that reason, ambiguity handling should be treated as a decision problem about *what to do next* under uncertainty, not only as a parsing problem.

Existing robotic natural language understanding systems handle several ambiguity forms in isolation, including indirect speech acts, clarification requests, and fuzzy parameter resolution [Zhang *et al.* 2025; Madureira and Schlangen 2023; Sucker *et al.* 2024; Sucker and Henrich 2025]. For example, Indirect Speech Act (ISA) commands such as “*Can you pass that tool?*” require pragmatic reinterpretation, while fuzzy commands such as “*Prepare some food in about ten minutes!*” and quantity ranges such as “*three to eight arbitrary drinks*” require parameter grounding [Zhang *et al.* 2025; Scheutz *et al.* 2022; Sucker *et al.* 2024; Sucker and Henrich 2025]. However, these mechanisms are typically implemented as separate modules and do not adequately coordinate resolution when multiple ambiguity types co-occur in a single command. Existing meta-level routing approaches are also typically driven by ambiguity degree thresholds rather than explicit ambiguity-type and task-risk classification [Zhou *et al.* 2026].

The project evaluates a specific coordination problem: whether ambiguity-type and risk-aware routing decisions are more appropriate than direct parsing or uniform handling strategies for compound ambiguous robot commands. When pragmatic, referential, and temporal uncertainties combine, as in “*Can you move that thing over there in a few minutes?*”, uncoordinated pipelines can produce repeated clarification loops or unsafe silent assumptions because independently designed modules make incompatible assumptions [Sarathy *et al.* 2025; Madureira and Schlangen 2023; Sucker *et al.* 2024; Jackson 2022].

Research question. *Does a risk-aware ambiguity manager improve routing decisions for compound ambiguous robot commands compared to direct Large Language Model (LLM) interpretation and uniform ambiguity-handling baselines, when evaluated using ambiguity classification accuracy, routing correctness, clarification appropriateness, and intent/parameter correctness on a labelled hybrid command benchmark?*

Research aim and objectives. This project aims to design and evaluate a routing framework that classifies ambiguity type and task risk in robot commands, then selects an appropriate resolution strategy. To achieve this, the project has the following objectives:

1. Define an ambiguity taxonomy for robot command interpretation that aligns with the methodology label schema.
2. Construct and harmonise a labelled hybrid command benchmark from AmbiK, the SaGC dataset, TEACH/TEACH-DA, IndirectRequests, and a controlled manual compound-ambiguity extension validated for ambiguity diversity and realism [Ivanova *et al.* 2025; Park *et al.* 2024; Padmakumar *et al.* 2022; Mannekote *et al.* 2024].
3. Implement representative baselines: direct LLM interpretation, always clarify (a highly conservative policy), always silently resolve (a maximum-efficiency policy), and degree-based routing [Zhou *et al.* 2026].
4. Implement a type-and-risk-aware routing manager that selects strategy decisions (execute, clarify, silently_resolve, face_preserving_rejection, multi_step).
5. Evaluate all systems using routing correctness as the primary metric, with supporting ambiguity, clarification, intent and Core Propositional Content (CPC) correctness, and risk-sensitive metrics.
6. Analyse failures by ambiguity type, risk level, source dataset, and compound-ambiguity status, and identify limitations for future embodied deployment.

Hypothesis. A routing layer that explicitly classifies ambiguity type and task risk before choosing a resolution strategy will yield higher routing correctness, specifically in clarification, face-preserving rejection, and silently_resolve decisions, than systems that rely on a single uniform ambiguity-handling strategy.

Expected contribution. The project contributes:

- an explicit ambiguity taxonomy for routing-level command interpretation;
- a labelled hybrid command benchmark focused on routing decisions, including controlled compound-ambiguity examples;
- a risk-aware routing framework that coordinates existing ambiguity-handling strategies rather than replacing all downstream robot execution components;
- an empirical comparison against direct LLM interpretation and uniform ambiguity-handling baselines using shared metrics and failure analysis.

The project focuses on classification and routing evaluation in the natural language understanding (NLU) coordination layer rather than full physical robot execution.

Proposal overview. Chapter 2 reviews related work on robot command interpretation, indirect speech acts, clarification and ambiguity detection, fuzzy parameter grounding, and risk-aware rejection, then identifies the gap this study targets. Chapter 3 presents the methodology, including the proposed type-and-risk-aware manager, hybrid dataset strategy, annotation schema, baselines, metrics, experimental procedure, and failure-analysis plan. Chapter 4 presents the project schedule. Chapter 5 closes by summarising the problem, approach, and expected contribution.

Chapter 2

Background and Related Work

2.1 Natural Language Command Interpretation in Robotics

Natural-language robot interfaces improve accessibility, but they also expose interpretation pipelines to ambiguity that can propagate into incorrect or unsafe behaviour if it is not managed explicitly [Zhang *et al.* 2025]. Prior work has improved parsing and grounding components, including symbolic-plus-LLM pipelines and direct LLM parsing, but these systems are usually evaluated as interpretation modules rather than as explicit strategy routers for downstream ambiguity handling [Sarathy *et al.* 2025].

That distinction matters here. The work does not test whether language can be parsed at all; it tests whether routing decisions improve when ambiguity is classified by semantic type and task risk before action selection. Integrated pipelines exist (for example, integrated architectures that combine pragmatic interpretation, reference resolution, and norm/capability checks), but current evidence still points to the need for stronger explicit coordination when multiple ambiguity types interact in compound cases [Sarathy *et al.* 2025].

2.2 Large Language Models for Structured Robotic Outputs

For routing-oriented robot command interpretation, LLMs are most useful when they are constrained to produce machine-actionable outputs rather than free-form text [Sarathy *et al.* 2025; Zhou *et al.* 2026]. In practice, this means decoding into a fixed schema (for example, JSON fields for intent, slots, ambiguity labels, risk level, and recommended strategy), so outputs can be consumed directly by downstream modules with predictable structure [Zhou *et al.* 2026].

This also motivates the prompting-versus-adaptation distinction. Prompt-only approaches can work, but ambiguity-heavy settings often benefit from task-specific adaptation and stricter output constraints when the system must follow a fixed decision protocol across edge cases [Park *et al.* 2024; Zhou *et al.* 2026; Sarathy *et al.* 2025].

2.3 Indirect Speech Acts (ISA) in Robot Commands

Indirect speech acts introduce a pragmatic mismatch between surface form and intended action, so literal parsing alone can route commands incorrectly even when lexical content is well-formed [Zhang *et al.* 2025]. Foundational speech-act and pragmatics theories suggest that intended force is often inferred from context and cooperative norms rather than literal syntax alone [Austin 1962; Searle 1969 1975; Grice 1975]. In robot settings, this is more than a semantic issue: ISA misinterpretation can degrade teaming outcomes and produce brittle execution choices when directive intent is not recovered before planning, including measured effects on trust and goal alignment in physical collaboration [Zhang *et al.* 2025; Sarathy *et al.* 2025].

Related dialogue data work also indicates that natural task-oriented commands include indirect user requests that require pragmatic inference rather than direct slot filling, reinforcing ISA handling as a routing-relevant ambiguity type [Mannekote *et al.* 2024].

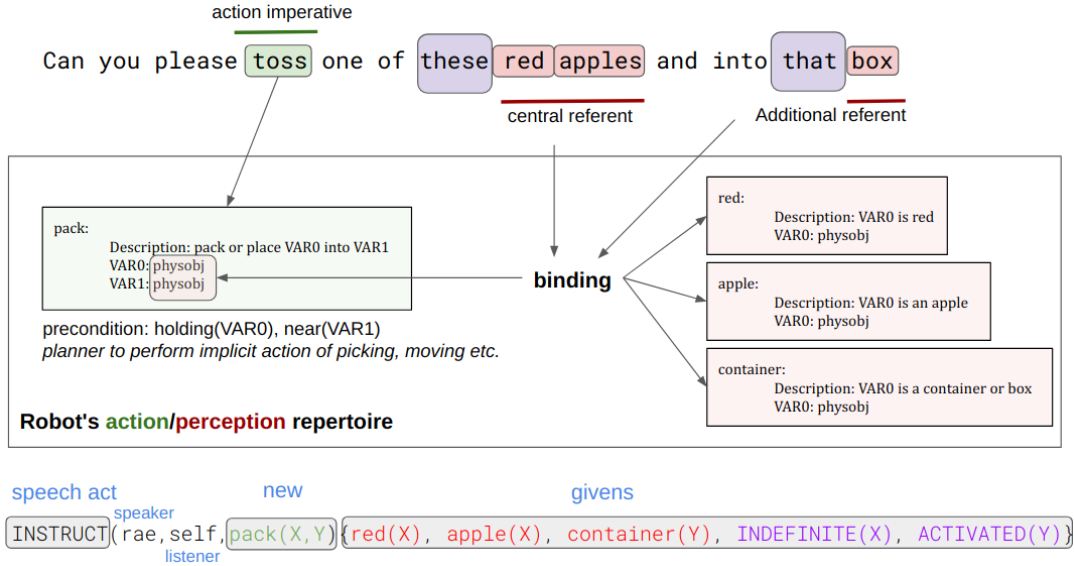


Figure 2.1: Interpreting a natural-language imperative requires identifying the speech act (intent), core propositional content (new information), and supplemental propositional content (given knowledge used for reference resolution). Adapted from Sarathy *et al.* [2025].

For this project, ISA handling motivates a dedicated ambiguity label and an explicit routing decision point. Pragmatically indirect commands may need reinterpretation before any clarification or parameter-grounding step is meaningful; otherwise later modules can operate on the wrong intent structure and compound failure risk [Sarathy *et al.* 2025]. This supports a sequencing view where ambiguity types are handled separately first, then coordinated through routing.

2.4 Clarification and Ambiguity Detection

Clarification literature shows that deciding *when* to ask is a separate problem from deciding *what* a command means, and that this decision can be modeled either as explicit prediction or as thresholded escalation from internal resolution [Madureira and Schlangen 2023; Zhou et al. 2026]. These methods demonstrate that clarification appropriateness is measurable, but they are usually optimized around underspecification severity and local confidence rather than a broader ambiguity-type portfolio. Embodied dialogue benchmarks such as TEACH and DialFRED further motivate studying clarification in multi-turn task contexts rather than single-turn utterances alone [Padmakumar et al. 2022; Gao et al. 2022].

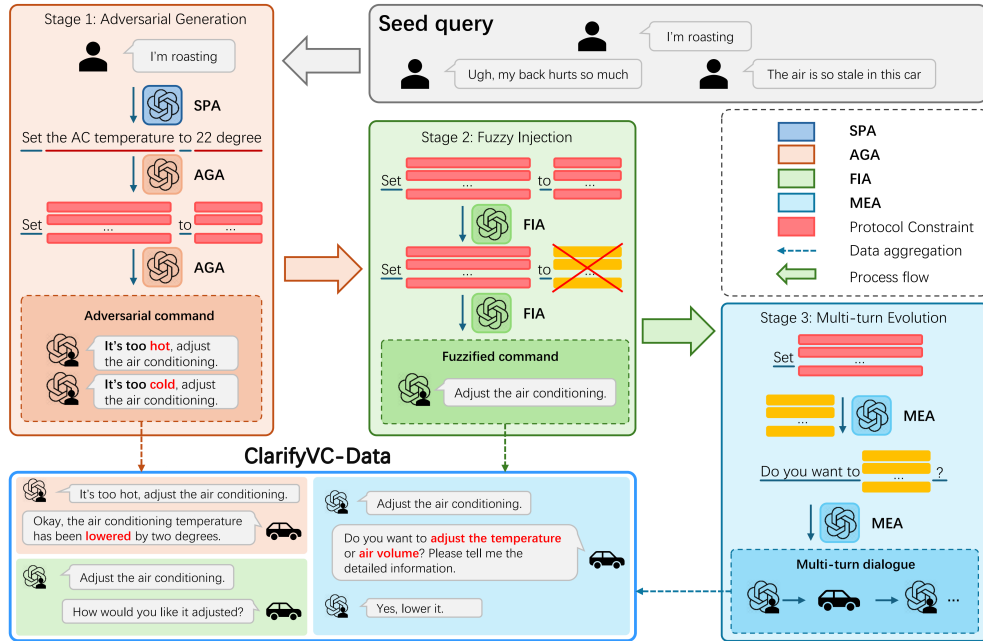


Figure 2.2: The ClarifyVC stage-wise generation flow. The schema-constrained pipeline combines semantic parsing, adversarial construction, and multi-turn evolution to resolve ambiguous commands. Adapted from Zhou et al. [2026].

Work on morally sensitive interactions also shows that clarification and rejection are coupled: reflexive clarification without adequate ethical filtering can itself be undesirable in high-stakes requests [Jackson 2022]. This supports the proposal's focus on routing decisions that combine ambiguity interpretation with risk and capability context.

That limitation is central to the proposal gap. A severity-first policy can improve detection for some fuzzy commands, yet commands with similar apparent ambiguity degree can require different actions (e.g., reinterpretation for ISA vs. clarification for missing referent vs. rejection for infeasibility) [Madureira and Schlangen 2023; Zhou et al. 2026]. ClarifyVC also defines ambiguity categories, but its operational decision boundary is still largely severity-oriented for escalation, which motivates testing whether

explicit type-and-risk coordination improves routing on compound cases [Zhou *et al.* 2026].

2.5 Fuzzy Temporal and Quantitative Parameters

Fuzzy temporal and quantitative work shows that many vague commands can be resolved through formal internal grounding without immediate user interruption, using satisfaction-based temporal models or globally constrained quantity grounding [Sucker *et al.* 2024; Sucker and Henrich 2025]. These methods provide a technical basis for the `silently_resolve` route; in this proposal, that route is restricted to cases where task risk is low enough for internal resolution to be acceptable. Compared with other ambiguity categories, this area currently has relatively concentrated formal treatment in the reviewed sources [Sucker *et al.* 2024; Sucker and Henrich 2025].

However, these methods are generally evaluated within their own ambiguity families and assume upstream interpretation is already correct. They provide limited guidance for cross-type coordination when fuzzy parameters co-occur with pragmatic or referential uncertainty in one command [Sucker *et al.* 2024; Sucker and Henrich 2025; Sarathy *et al.* 2025]. This directly supports the proposal’s compound-ambiguity focus and the need to test routing behaviour on multi-label cases rather than single-type examples only.

2.6 Risk, Capability, and Transparent Rejection

Risk- and capability-aware work indicates that some commands should not be clarified or silently resolved; they should be transparently rejected when authorization, feasibility, or normative constraints fail [Scheutz *et al.* 2022; Park *et al.* 2024; Jackson 2022]. This establishes rejection as a first-class strategy in command interpretation, not just an execution-time exception.

Recent routing-style datasets and safety-oriented benchmarks further motivate separating infeasible or unsafe cases from merely ambiguous ones, because the communicative objective changes from “recover missing information” to “justify non-execution” [Park *et al.* 2024; Yin *et al.* 2024]. For this proposal, that evidence motivates explicit risk and capability fields in the shared schema and risk-sensitive metrics such as safe rejection and unsafe silent-resolution rates. Uncertainty signaling is also central to this decision. Prior routing work highlights two practical mechanisms: predictive-token uncertainty when model probabilities are available, and context-sampling variance when they are not [Park *et al.* 2024]. The context-sampling approach repeatedly queries the model with controlled context permutations and measures output variance, providing a model-agnostic uncertainty proxy for routing decisions [Park *et al.* 2024].

2.7 Synthesis and Research Gap

The literature supports a clear synthesis claim: robot command interpretation has credible mechanisms for multiple ambiguity categories, but these mechanisms are mostly developed and evaluated in isolation rather than as a coordinated decision process for compound commands [Sarathy *et al.* 2025; Madureira and Schlangen 2023; Sucker and Henrich 2025]. Existing routing approaches provide useful threshold policies, yet they are predominantly degree-based and do not fully operationalize type-specific coordination across ISA, clarification, fuzzy grounding, and rejection under risk [Zhou *et al.* 2026]. The literature implicitly defines a hierarchical dependency between ambiguity types (pragmatic $\rightarrow$ clarification $\rightarrow$ parametric), and existing systems partially operationalise aspects of this hierarchy, but do not explicitly coordinate ambiguity types and task risk in a unified routing framework. Existing evaluations focus on isolated metrics rather than coordinated decision quality under compound ambiguity, motivating routing correctness as the primary evaluation target.

This frames the project directly. The open question is not whether each ambiguity mechanism can work in isolation, but whether a type-and-risk-aware manager improves routing decisions over direct LLM interpretation and uniform baselines when ambiguity types co-occur [Sarathy *et al.* 2025; Park *et al.* 2024]. Dataset evidence shows progress on ambiguity categorisation (for example, AmbiK’s ambiguity groupings), but still not full coverage of the compound routing problem targeted here [Ivanova *et al.* 2025]. The methodology addresses this by treating routing as the primary outcome and using a hybrid benchmark (AmbiK, SaGC, TEACH/TEACH-DA, IndirectRequests, plus a controlled compound extension) with failure analysis by ambiguity type, risk level, and compound status [Ivanova *et al.* 2025; Park *et al.* 2024; Padmakumar *et al.* 2022; Mannekote *et al.* 2024]. By combining information-structure decomposition (Figure 2.1) with stage-wise ambiguity-routing logic (Figure 2.2), the next chapter specifies a methodology to evaluate unified type-and-risk-aware routing against uniform baselines on compound ambiguity cases. As summarised in Table 2.1, the literature highlights a clear divide between current isolated capabilities and the missing meta-level coordination required for compound ambiguity.

Current capability	Missing capability
Resolve fuzzy temporal parameters via satisfaction functions	Select resolution strategy based on ambiguity type, not just degree
Resolve fuzzy quantitative parameters via global grounding	Handle compound co-occurring ambiguity types in a single utterance
Detect underspecification and generate targeted clarification requests	Classify pragmatic, referential, and parametric ambiguity before acting
Comprehend indirect speech acts and extract illocutionary intent	Route between ISA resolution, clarification, and silent resolution dynamically
Reject unsafe commands transparently with justification	Model physical task stakes in routing decisions

Table 2.1: Summary of current capabilities and identified research gaps across the surveyed literature (ISA: Indirect Speech Act).

Chapter 3

Research Methodology

3.1 Research Design

The study uses a quantitative experimental design. The central comparison is whether a type-and-risk-aware routing policy improves ambiguity-handling decisions compared with direct LLM interpretation and uniform baselines on a labelled hybrid benchmark.

The research scope is intentionally limited to the natural-language coordination layer: ambiguity classification, risk/capability assessment, and routing strategy selection. This isolates the decision component under study and avoids confounding the evaluation with perception, motion planning, and actuator-level failures. This scope is consistent with prior evaluation approaches that separate interpretation quality from downstream physical execution variability [Sarathy *et al.* 2025; Zhou *et al.* 2026].

3.2 Proposed Risk-Aware Ambiguity Manager

The proposed method is a meta-level routing manager that coordinates ambiguity-handling strategies rather than replacing all downstream robot subsystems. The manager will be implemented as a supervised fine-tuned open-source LLM on the shared hybrid benchmark, using schema-constrained decoding so each prediction is emitted as structured JSON with ambiguity labels, risk/capability fields, and a routing decision [Park *et al.* 2024; Zhou *et al.* 2026; Sarathy *et al.* 2025]. For each command, the manager consumes:

- raw natural-language command;
- optional dialogue history;
- optional scene/context state;
- optional capability context.

It produces:

- inferred intent and slot-level interpretation;
- ambiguity labels (single or multi-label);
- risk level and capability status;
- recommended strategy;
- optional clarification question or rejection explanation.

The routing strategy set is:

- `execute`
- `clarify`
- `silently_resolve`
- `face_preserving_rejection`
- `multi_step`

Strategy semantics are fixed during annotation and evaluation: `execute` routes a clear, safe command directly to execution; `clarify` asks a targeted question to recover missing critical information; `silently_resolve` internally grounds low-risk fuzzy parameters without interrupting the user; `face_preserving_rejection` refuses unsafe or infeasible commands with a brief transparent explanation; and `multi_step` applies an ordered strategy sequence for compound ambiguity. The term `face_preserving_rejection` is used because the rejection response is designed to protect user trust and social dignity while still communicating clear safety or feasibility constraints.

The manager treats compound ambiguity as a coordination problem. It identifies ambiguity type and stakes first, then routes to an action policy. This framing follows prior work that treats indirect speech acts as pragmatic intent-recovery problems, clarification as a separate decision process, fuzzy temporal/quantitative language as parameter-grounding problems, and infeasible or unsafe instructions as explicit rejection or constraint-handling problems [Zhang *et al.* 2025; Madureira and Schlangen 2023; Zhou *et al.* 2026; Sucker *et al.* 2024; Sucker and Henrich 2025; Scheutz *et al.* 2022]. For uncertainty estimation, the manager will use context sampling with output-variance scoring: the same command is evaluated across controlled permutations of contextual input, and prediction instability is used as an uncertainty feature for escalation and risk-sensitive routing [Park *et al.* 2024]. Figure 3.1 summarises the proposed routing flow.

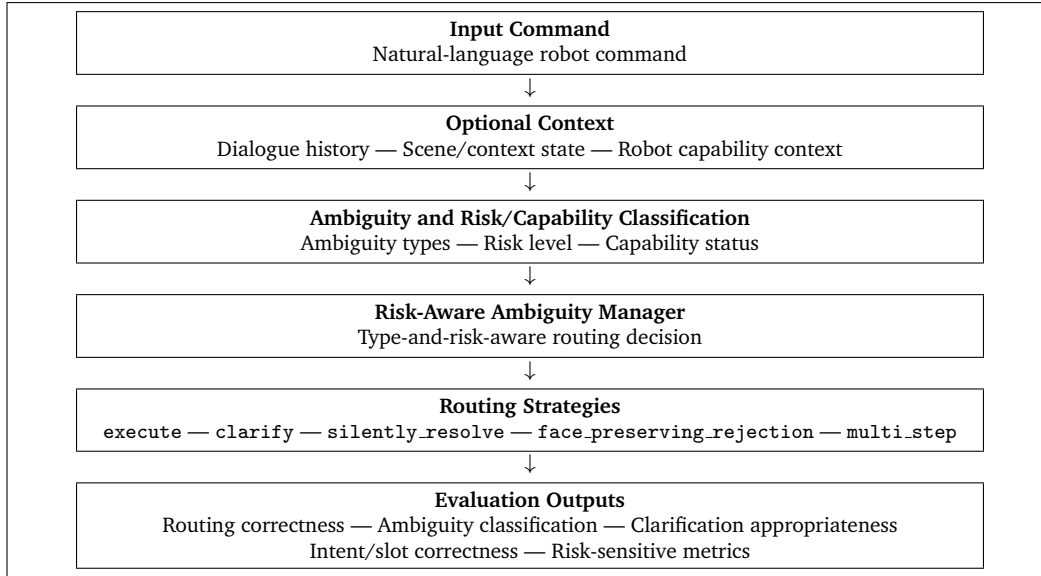


Figure 3.1: Proposed risk-aware ambiguity manager. The manager classifies ambiguity type, task risk, and capability status before routing a command to an appropriate handling strategy. Evaluation checks whether the selected route and structured interpretation match the labelled benchmark.

3.3 Dataset Strategy

The evaluation uses a hybrid benchmark built from datasets that cover different parts of the routing problem. Representative source-backed examples include ISA commands (“Can you pass that tool?”), fuzzy temporal commands (“Prepare some food in about ten minutes!”), flexible quantity commands (“three to eight arbitrary drinks”, “one to four blue gears”), and compound ambiguity commands (“Can you move that thing over there in a few minutes?”) [Zhang *et al.* 2025; Sucker *et al.* 2024; Sucker and Henrich 2025; Sarathy *et al.* 2025].

- **AmbiK**: ambiguity-focused robot tasks with ambiguous/unambiguous pairs and clarification-relevant supervision [Ivanova *et al.* 2025].
- **SaGC dataset**: high-level goals with scene context and clear/ambiguous/infeasible labels for routing evaluation (introduced with the CLARA framework) [Park *et al.* 2024].
- **TEAch/TEAch-DA**: embodied, multi-turn household dialogue with question-and-answer behaviour and cooperative interaction that supports the study of mistake recovery [Padmakumar *et al.* 2022].
- **IndirectRequests**: controlled indirect user requests for evaluating indirect speech act handling [Mannekote *et al.* 2024].
- **Manual compound extension**: a controlled set of 50 targeted compound ambiguity examples for combinations underrepresented in public datasets.

No single source contains all labels required by this project’s taxonomy. A hybrid benchmark is therefore necessary because coverage is distributed across sources. To keep the work feasible within the honours timeline, the benchmark will use a controlled subset of approximately 250–400 examples selected from the core datasets, together with a manually constructed extension of 50 targeted compound-ambiguity examples. The project does not attempt exhaustive annotation of all source corpora; instead, examples will be selected to balance source dataset, ambiguity type, risk level, and compound-ambiguity status where possible.

SafeAgentBench is included as a targeted safety add-on for stress-testing rejection behaviour [Yin *et al.* 2024]. Dynamic-RDMM and ReferIt3D/RefCOCO/Collaborative Manipulation Corpus are reserved for augmentation and separate referential-grounding sub-evaluations. The hybrid framing is consistent with reports that no single benchmark fully covers the compound ambiguity categories and safety-sensitive routing decisions required by this study.

3.4 Annotation Schema

All source examples will be converted into a shared JSONL schema for consistent training/evaluation interfaces. This schema is proposed for this project and is not presented as complete in each original dataset. Core fields are:

- gold_intent
- gold_slots
- ambiguity_present
- ambiguity_types
- primary_ambiguity_type
- compound_ambiguity
- compound_ambiguity_count
- risk_level
- risk_target
- capability_status
- recommended_strategy
- strategy_sequence
- gold_clarification_question
- gold_reinterpretation
- gold_explanation
- success_condition

Several fields (especially risk-level targets, compound ambiguity labels/counts, and strategy sequences) are not expected to be natively available across all source datasets and will require explicit manual annotation under written guidelines. The methodology therefore treats schema completion as a substantial annotation task, not a trivial automatic mapping step. Mapping and annotation rules will be documented so provenance is preserved and cross-dataset comparisons remain valid. Annotation decisions will be calibrated on concrete command types: ISA (“*Can you pass that tool?*”), fuzzy temporal (“*Prepare some food in about ten minutes!*”), fuzzy quantity (“*three to eight arbitrary drinks*”), and compound ambiguity (“*Can you move that thing over there in a few minutes?*”). To mitigate subjectivity in manual annotations, a rule-based annotation guideline will be defined for ambiguity type, compound ambiguity identification, and task risk classification. Where feasible, a subset of the dataset will be double-annotated to assess inter-annotator agreement, and disagreements will be resolved through predefined criteria. Inter-annotator agreement will be measured using metrics such as Cohen’s kappa [Cohen 1960]. To improve reproducibility, the guideline will define each label with operational criteria rather than only intuitive descriptions. For example, ambiguity-type definitions will specify the minimum evidence required for assigning pragmatic, referential, temporal, or quantitative categories, and will include short borderline examples where multiple labels may be plausible. Compound-ambiguity annotation will require explicit justification for why a case is multi-label rather than a single dominant ambiguity. Source provenance will be preserved for each example so that evaluation can be stratified by originating dataset and annotation pathway. In addition, annotators will document concise reasons for assigned risk and recommended strategy labels, enabling transparent review of difficult cases and more consistent disagreement resolution.

3.5 Baseline Systems

Five systems will be compared:

1. **Direct LLM interpretation:** evaluates whether explicit routing logic is needed; this corresponds to direct parser-style LLM command interpretation without a dedicated coordination layer [Sarathy et al. 2025].
2. **Always clarify (maximum conservatism):** tests over-clarification and interaction cost.
3. **Always silently resolve (maximum efficiency):** tests failure modes from aggressive non-interruption, including unsafe guesses and brittle resolution behaviour under severe ambiguity.
4. **Degree-based routing:** severity-threshold policy without explicit ambiguity-type semantics, motivated by clarification-threshold style routing where stronger fuzziness triggers counter-questioning [Zhou et al. 2026].
5. **Type-and-risk-aware routing (proposed):** strategy chosen using ambiguity type, risk, and capability together.

This baseline set spans heuristic policies and one threshold-style comparator, which helps attribute gains to explicit coordination rather than implementation detail alone. Clarification-first behaviour is motivated by clarification-decision literature [Madureira and Schlangen 2023], while silent-resolution behaviour is motivated by fuzzy-parameter grounding literature [Sucker *et al.* 2024; Sucker and Henrich 2025]. While always-clarify and always-resolve represent intentionally extreme policies, they are included to establish upper and lower behavioural bounds, with degree-based routing providing a stronger and more realistic comparison. The design isolates the marginal benefit of type-and-risk-aware routing over direct parsing, threshold-only routing, and single-policy heuristics.

3.6 Evaluation Metrics

The primary system-level metric is **routing correctness**, a project-defined measure of whether the meta-manager selects the appropriate downstream handling path. It is defined as the proportion of examples where predicted strategy matches gold recommended strategy, and adapts strategy-selection ideas used in ambiguity-routing work (for example, CLARA-style clear/ambiguous/infeasible classification and ClarifyVC escalation decisions) [Park *et al.* 2024; Zhou *et al.* 2026].

Secondary metrics include:

- ambiguity classification micro-F1 and macro-F1;
- exact ambiguity-set match;
- clarification precision and clarification recall;
- intent correctness;
- slot binding correctness;
- risk-sensitive decision accuracy;
- safe rejection rate;
- unsafe silent-resolution rate.

$$\text{Routing Correctness} = \frac{|\{i : \hat{r}_i = r_i\}|}{N}$$

$$\text{Clarification Precision} = \frac{\text{TP}_{\text{clarify}}}{\text{TP}_{\text{clarify}} + \text{FP}_{\text{clarify}}}$$

$$\text{Clarification Recall} = \frac{\text{TP}_{\text{clarify}}}{\text{TP}_{\text{clarify}} + \text{FN}_{\text{clarify}}}$$

$$\text{Clarification F1} = 2 \times \frac{\text{Clarification Precision} \times \text{Clarification Recall}}{\text{Clarification Precision} + \text{Clarification Recall}}$$

$$\text{Unsafe Silent-Resolution Rate} = \frac{|\{i : \hat{r}_i = \text{silently_resolve} \wedge i \in \text{high-risk/unsafe set}\}|}{|\{i : i \in \text{high-risk/unsafe set}\}|}$$

Here, $\hat{r}_i$ is the predicted routing strategy, r_i is the gold routing strategy, and N is the number of evaluation examples. TP , FP , and FN are computed with respect to the clarify routing decision.

Reported results will include aggregate values and subgroup breakdowns by ambiguity type, risk level, source dataset, and compound-ambiguity status. The Clarification F1 score will serve as the primary comparative metric against baseline systems such as CLARA, which use F1 to evaluate whether the robot asks questions only when ambiguity is present. Clarification precision/recall are grounded in clarification-prediction evaluation practice [Madureira and Schlangen 2023]. Intent and structure-level correctness are motivated by information-structure evaluation framing for robot command interpretation [Sarathy et al. 2025]. Safety-oriented rejection metrics are motivated by transparent rejection/safety handling work and safety benchmark framing [Scheutz et al. 2022; Yin et al. 2024].

3.7 Experimental Procedure

The procedure is:

1. Obtain and inspect core datasets.
2. Convert data into the shared schema.
3. Construct manual compound-ambiguity extension examples.
4. Create source-balanced development/evaluation partitions.
5. Implement all baselines.
6. Implement the proposed manager.
7. Run all systems on the same held-out evaluation set.
8. Compute primary and secondary metrics.
9. Conduct structured failure analysis.

All systems will be evaluated under the same schema and metric definitions so the comparisons are fair.

3.8 Failure Analysis

Failure analysis will identify where and why decisions fail. Errors will be grouped by:

- ambiguity type;
- compound versus non-compound ambiguity;
- risk level;
- source dataset;
- selected strategy;
- baseline category.

Special attention will be paid to unsafe silent-resolution and over-clarification patterns, including ISA misrouting on “*Can you pass that tool?*” and compound-case breakdowns on “*Can you move that thing over there in a few minutes?*”, and to whether failures arise from upstream ambiguity/risk classification or downstream routing choice. This failure-breakdown style is consistent with prior ambiguity-focused evaluation that separates detection, interpretation, and decision stages [Madureira and Schlangen 2023; Sarathy et al. 2025]. Concretely, analysis will separate at least three error layers: ambiguity-classification errors (wrong or incomplete labels), risk-classification errors (misestimated stakes), and routing errors (incorrect strategy choice given predicted labels). This decomposition helps identify whether performance limits come from representation quality or policy quality. Results will also be compared across baselines to identify baseline-specific failure patterns, such as over-clarification in conservative policies or unsafe guesses in aggressive silent-resolution policies. Even if the proposed manager does not outperform every baseline on every metric, these diagnostics are still useful because they show which decision components generalise and which need refinement in later routing-policy iterations.

3.9 Limitations and Ethical Considerations

This methodology is limited to text-first routing evaluation and does not directly measure embodied execution effects. Additional limitations include distributed dataset assumptions, partial reliance on manually constructed compound examples, and possible subjectivity in risk/capability labels when labels are inferred rather than provided.

No human-subject study is planned in the initial phase, so user-facing quality outcomes are evaluated indirectly via routing and interpretation metrics. Public datasets will be used according to licence terms, and manual examples will exclude sensitive personal content. The ethical motivation is to reduce unsafe behaviour through explicit clarification/rejection for risky or infeasible commands. This safety communication framing is consistent with prior work on transparent rejection and justification in task-based human-robot communication [Scheutz et al. 2022].

Chapter 4

Schedule of Work

The project will be completed during Semester 2 (August–October) in phased milestones with explicit dependencies so that later evaluation tasks depend on completed data and system components. Table 4.1 summarises the schedule.

Weeks	Phase	Planned Work (with dependency)
1–2 (Aug)	Proposal finalisation	Finalise scope, evaluation definitions, and annotation guidelines aligned with Chapters 2–3.
3–4 (Aug)	Dataset construction	Inspect AmbiK, SaGC, TEACH/TEACH-DA, and IndirectRequests; map selected examples to the shared schema; set up initial annotation on a controlled subset; prepare quality checks. <i>Required before evaluation.</i>
5 (early Sep)	Manual extension	Construct and validate 50 targeted compound-ambiguity examples, following completion of the initial schema mapping; continue annotation refinement and validation.
6–7 (Sep)	Baseline implementation	Implement direct LLM, always clarify, always silently resolve, and degree-based routing baselines. <i>These baselines must be stable before implementing and comparing the manager.</i>
8–9 (late Sep)	Manager implementation	Implement the type-and-risk-aware routing manager and integrate ambiguity-type, risk, and capability signals after baseline interfaces are fixed.
10–11 (Oct)	Evaluation	After completion of dataset preparation and all systems, run held-out evaluation and compute routing, ambiguity, clarification, intent/slot, and risk-sensitive metrics; complete final annotation validation checks.
12 (late Oct)	Analysis and writing	Following evaluation, perform failure analysis by ambiguity type/risk/source/compound status, then complete final write-up and revision.

Table 4.1: Semester 2 (August–October) schedule for implementation, evaluation, and write-up.

The schedule is feasible for an honours project because it limits scope to routing-layer evaluation, reuses public datasets, and keeps dependency ordering clear from

dataset construction through baselines, manager implementation, evaluation, and final analysis/write-up. The staged order is practical because each phase produces artifacts needed by the next phase: a controlled subset and initial schema mapping enable baseline implementation, and stable baseline interfaces support manager integration before held-out evaluation. Dataset subset construction is completed before full comparative evaluation to avoid moving-target metrics during system comparison. Annotation refinement and validation continue alongside development so category definitions and risk labels stay consistent as edge cases appear. The final weeks are reserved for structured failure analysis, interpretation, and writing, which reduces the risk of unfinished analysis at submission time.

Chapter 5

Conclusion

Ambiguous natural-language robot commands are still hard to handle reliably when different ambiguity types co-occur and require different response strategies.

The proposal addresses that gap by introducing a type-and-risk-aware ambiguity manager that classifies ambiguity and task risk, then routes commands to execute, clarify, silently resolve, face-preserving rejection, or multi-step handling. The evaluation plan uses a labelled hybrid benchmark and compares the manager against direct LLM interpretation and uniform handling baselines, with routing correctness as the primary metric and ambiguity, clarification, intent/slot, and risk-sensitive measures as supporting evidence.

The expected contribution is a practical routing-focused framework and empirical evidence on whether explicit ambiguity-type and risk-aware coordination improves decision quality for ambiguous robot commands within a feasible honours-scale scope. Feasibility is supported by the routing-only scope, controlled-subset benchmark design, and reuse of shared baselines and metrics across all systems. These constraints keep implementation and analysis bounded while preserving a meaningful empirical contribution that can still be evaluated rigorously within honours project limits.

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